Bit Manipulation

$\boxed{\text{Number system}}$

Decimal number

digits

\# symbols = 10

0 1 2 … 9

Face Value

8 4 2 1

hundreds

ones

tens

thousands

Place value

4281

$$8 \times 1000 + 4 \times 100 + 2 \times 10 + 1 \times 1$$

$$8000 + 400 + 20 + 1$$

x → 0

m → 1

\# → 2

^

~

B

n/p

10

10

$$\text{Value} = \sum_{\text{digit}} \text{face value} \times \text{place value}$$

Decimal Num System place value

7 8 4 2 1

10^4 10^3 10^2 10^1 10^0

Obs B

place value inc by a power of 10

pv (i^{th} digit from the right) $= 10^{i-1}$

$i = 1, 2, 3, 4, 5, 6, 7, 8$

Obs A : 10 symbols

obs B : Place value inc by a power of 10

III IV

V

VI ...

X

1

5

9

$\boxed{1\,0}$

1

99

100

<u>Base</u> = 4

$\boxed{\begin{array}{c}0\\1\\2\\3\end{array}}$

0

3

0_4

1_4

2_4

3_4

10_4

11_4

4^0

$1\times4^1 + 1\times4^0 = 4+1 = 5$

4^1

$1 \times 4^1 + 0 \times 4^0$

$= 4 + 0$

$= 4$

$\underset{8^3}{\_} \quad \underset{8^2}{0} \quad \underset{8^1}{\_} \quad \underset{8^0}{\_}$

$(8)^2 \times 0$

Octal [8 symbols]

0 1 2 3 4 5 6 7

$7 \times 8^1 + 3 \times 8^0$ $56 + 3 = 59$

$10_{10} \longrightarrow$

13_8 X

11_8 X

12_8 ☆

73_8 X

12_8

73_8 X

12_8

$1 \times 8^1 + 2 \times 8^0$

$8 + 2$

$= 10$

1 3

3×8^0

3

1×8^1

8 + 10

13_8 X

8 + 3 = 11

11_8

1×8^1 1×8^0

8 + 1

125_8

1×8^2 = 64 + 2×8^1 = 16 + 5×8^0 = 5

85

Any base to Decimal Conversion

$(471)_9$

$(4 \times 9^2 + 7 \times 9^1 + 1 \times 9^0)$

324 + 63 + 1

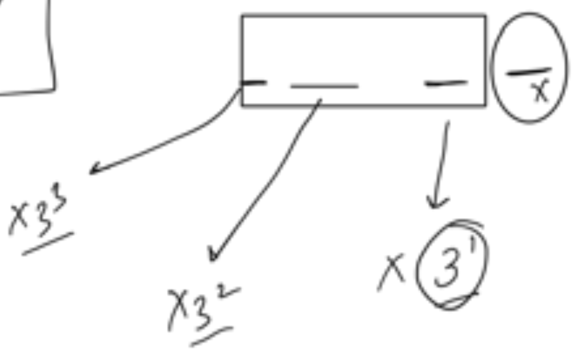
388_{10}

Decimal to Any base B

$42_{10} \longrightarrow \ ?_3$

Decimal   Ternary

Symbol
0
1
2



$3 \,\overline{)\, 42}$ → 14   divn
-3
12
-12
0   remainder

$3 \,\overline{)\, 14}$ → 4
-12
2

$3 \,\overline{)\, 4}$ → 1
-3
1

$3 \,\overline{)\, 1}$ → 0
-0
1

$(1\ 1\ 2\ 0)_3$

$42_{10} \equiv (1120)_3$

$1\times3^3 + 1\times3^2 + 2\times3^1 + 0\times3^0$

$27 + 9 + 6 + 0$

$36 + 6 = 42$

$90_{10} \equiv (?)_8$

$$
\begin{array}{r|l}
 & 11 \\
8 & 90 \\
 & -8 \\
 & 10 \\
 & -8 \\
 & 2
\end{array}
$$

$90_{10} \equiv ?_8$

$$
\begin{array}{r|l}
 & 11 \text{ quotient} \\
8 & 90 \\
 & -8 \\
 & 10 \\
 & -8 \\
 & 2 \text{ remain}
\end{array}
$$

$8\times11 + 2$

$(1\ 3\ 2)_8$

$8^4 \quad 8^3 \quad 8^2 \quad 8^1 \quad 8^0$

divisor * quotient + remainder = dividend

$$
\begin{array}{r|l}
 & 1 \\
8 & 11
\end{array}
\qquad
\begin{array}{r|l}
 & 0 \\
8 & 1
\end{array}
$$

$\frac{-8}{3}$

$\frac{-0}{1}$

$(6854)_8 \rightarrow$ 0

0 1 2 — 7

Quiz $(02101)_3$

$1\times3^2 + 0\times3^1 + 1\times3^0$

9 0 1

$+ 2\times3^3$

54

64

$\boxed{0 \quad 1 \quad \text{——} \quad 9}$

Base 11

A ≡ 10

B ≡ 11

C ≡ 12

D 13

E 14

F 15

Base 11

0 1 2 3 4 5 6 7 8 9 A

Hexadecimal = Base = 16

0 1 2 3 4 5 6 7 8 9 A

B

C

D

E

F

$(AC1)_{16}$

$1 = 1 \times 16^0$

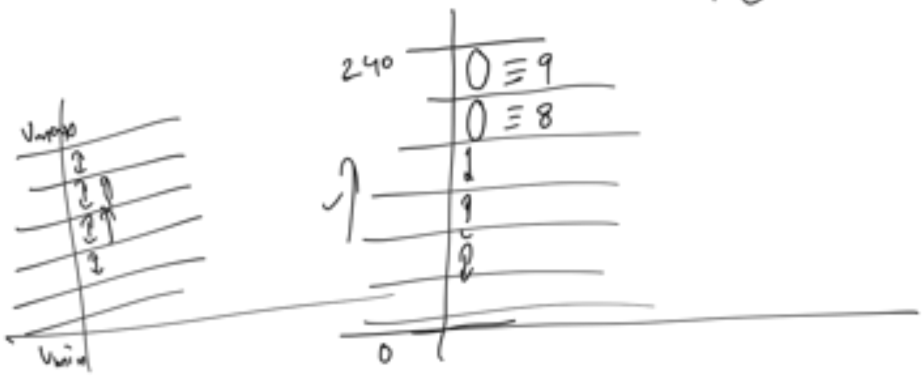
$192 = 12 \times 16^1$

$2560 = 10 \times 16^2$

2753

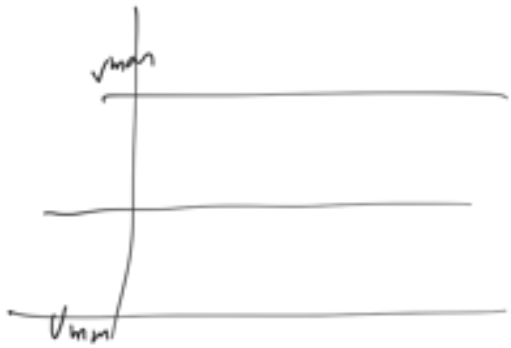
Class 10

Decimal = 10



Binary Number System

2



1 1 0 1 0 1 0 0 1

High
V
≡
ON

low
V
≡
OFF

$22_{10} \equiv (?)_2$

$$\begin{array}{r|l} 2 & 22 \\ & -22 \\ \hline & 0 \end{array} \quad 11$$

$$\begin{array}{r|l} 2 & 11 \\ & -10 \\ \hline & 1 \end{array} \quad 5$$

$$\begin{array}{r|l} 2 & 5 \\ & -4 \\ \hline & 1 \end{array} \quad 2$$

$$\begin{array}{r|l} 2 & 2 \\ & -2 \\ \hline & 0 \end{array} \quad 1$$

$$\begin{array}{r|l} 2 & 1 \\ & -0 \\ \hline & 1 \end{array} \quad 0$$

$(10110)_2$

Addition

Decimal

$$\begin{array}{r} {}^{1}\,{}^{1}\;\;{}^{1}\;\; \\ 1717 \\ 9815 \\ \hline 11532 \end{array}$$

$15 \% 10 = 5$

$1\cancel{7}/.0 = 2$

quotient = carry

$$\begin{array}{r} {}^{1}\,{}^{1}\,{}^{1}\;\; \\ 3459 \\ 2847 \\ \hline \boxed{\ }\boxed{\ }\boxed{0}\boxed{6} \end{array}$$

$$\begin{array}{r|l} 10 & 16 \\ & 10 \\ \hline & 6 \end{array} \quad 1$$

digit

$$\begin{array}{r|l} 10 & 10 \\ & 10 \\ \hline & 0 \end{array}$$

6 13 0 remainder

Addition in Binary

$$
\begin{array}{rl}
{}^{1\,1\,0} & \\
1\,0\,1\,1\,0 & A \\
0\,0\,1\,1\,1 & B \\
\hline
\boxed{1}\,\boxed{1}\,\boxed{1}\,\boxed{0}\,\boxed{1} & C
\end{array}
$$

C
11101

$2 \,|\, 2$, -2, 0 digit, 1 carry

$2 \,|\, 3$, 2, 1, 1

q: 47

q: 49

Bit Manipulation

Symbols ≡ Digit

[0, 1, 2, -- 9] ≡ 10 Symbols of Decimal Number System = Digit

≡

[0, 1] ≡ 2 Symbols of Binary Number System = Bit

Bitwise Operators

- & Bitwise AND
- | Bitwise OR
- ~ Bitwise NOT ≡ used on 1 operand
- ^ Bitwise XOR
- << >> Not today

10 + 20 = 30

−

×

÷

Logical Opr

True && True = T

True & false = F

Logical Operators

&& ≡ Logical AND

|| ≡ Logical OR

True

false

Boolean

Bitwise AND

```
  0 1 0 0 0 1 1
& 0 1 0 0 1 1 1
  -------------
  0 1 0 0 0 1 1
```

&		answer
0	0	0
0	1	0
1	0	0
1	1	1

```
111011
110010
------
110010
```

Bitwise OR

|		
0	0	0
0	1	1
1	0	1

if atleast 1 of the 2 inputs

```
  111011
| 110010
  ------
  111011
```

was

cout << 8 | 7 = 15

8 ≡ 1000
7 ≡ 0111
≡ 1111_2

1
+2
+4
+8
15

a = 13 ≡ 1101
b = 10 ≡ 1010
1000

a & b
= 8

print (13 & 10) = 8

10 ≡ 1010
1 ≡ 0001
1011
8 2 1 = 11

$\sim (10100)_2$

$$\begin{array}{r} \sim 10100 \\ \hline 01011 \end{array}$$

~ 4

$$\begin{array}{r} \sim 100 \\ \hline 011 = 3 \end{array}$$

$\sim 4 = 3$

---

^

Bitwise XOR

exclusive OR

$0 \wedge 0 = 0$

$0 \wedge 1 = 1$

$1 \wedge 0 = 1$

$1 \wedge 1 = 0$

exactly 1 out of 2 inputs == High

$A = 15 \equiv \boxed{1111}$

$B = 6 \equiv 0110$

print (A & B) = 0110 = 6

print (A | B) = 1111 = 15

print (A ^ B) = 1001 = 9

:)

Quick way of convert
D to B

A = 20

$$\begin{array}{r} 20 \\ -\ 16 \\ \hline 4 \end{array} \quad 4 - 4 = 0$$

1	0	1	0	0
2^4	2^3	2^2	2^1	2^0

A = 18 $\qquad$ 18 − 16 = 2 − 2 = 0

1 0 0 1 0

A = 250

$2^7 = 128$

$$\begin{array}{r} 250 \\ 128 \\ \hline 122 \\ 64 \\ \hline 58 \\ 32 \end{array}$$

$250_{10} \equiv$

1	1	1	1	1	0	1	0
7	6	5	4	3	2	1	0

26
16

10
-8

2
-2

0

A)) $\boxed{a \mid a = a}$

1 1 0 1 0 1 1 1
1 1 0 1 0 1 1 1

5 = 101
5 = 101
101

101
101
101

$\boxed{B \quad a \& a = a}$

C $\boxed{a \wedge a = \quad 0}$

101
^ 101

$a \& a = a$

$a | a = a$

$a \wedge a = 0$

$a \& 0 = 0$

$a | 0 = a$

$a \wedge 0 = a$

$a \& b == b \& a$

$a | b == b | a$

$a \wedge b == b \wedge a$

Order of operands do NOT matter ☺

$a \wedge b \wedge c \wedge d = b \wedge c \wedge a \wedge d = a \wedge d \wedge c \wedge b$

$a | b | c = b | c | a = c | a | b$

$10 \wedge 5 \wedge 10 = 10 \wedge 10 \wedge 5$

$$
\begin{array}{r}
10\ 10101 \\
11\ 10001 \\
11\ 00010 \\
\hline
10\ 00000
\end{array}
$$

A & B & C

$$
\begin{array}{lr}
A & 1010101 \\
| & \\
B & 1110001 \\
| & \\
C & 1100010 \\
\hline
 & 1110111
\end{array}
$$

A	1010101
∩	1110001
B	1100010
∩ C	
	1000110

For Bitwise AND ≡ all should be 1

For Bitwise OR ≡ atleast 1 bit should be 1

For Bitwise XOR ≡ odd # of bits should be 1

Q Stripe, Coinbase, Adobe, Amazon, MS

A = of size N where every number is present even number of times, but 1 number is present odd no of times

A = | 10 | 2 | 3 | 7 | 6 | 10 | 2 | 7 | 8 | 8 | 2 | 3 | 2 | 6 | 7 |

EV	#(10)	2	#(7)	3	Odd
EV	#(2)	4	#(8)	2	EV

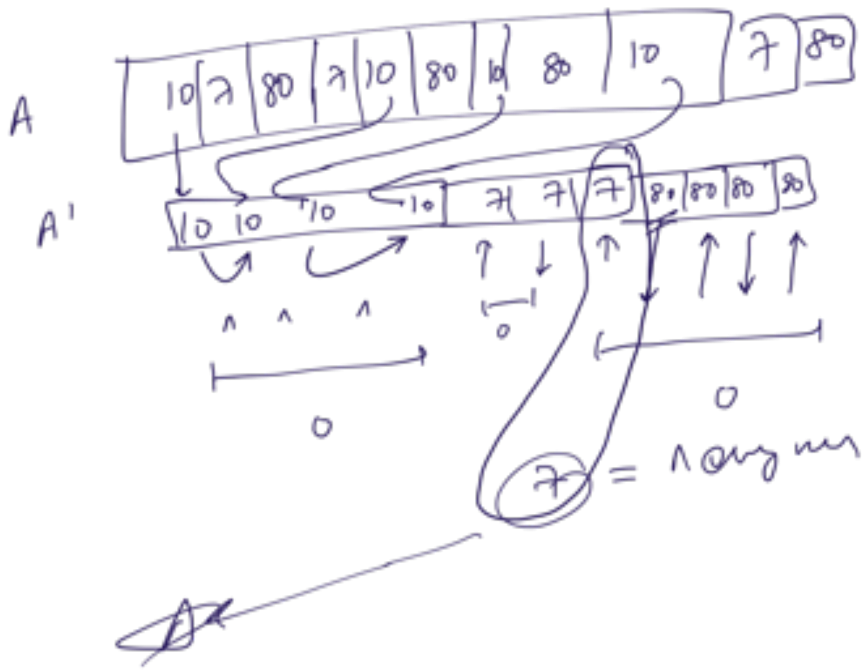
Gv #(3) 2

#(6) 2 kv

Q Find the number which was present odd number of times

7

^ ≡ Exclusivity




```
ans = a(0)
for(i=1, i<N, i++)
    ans = ans ^ a(i)
print(ans)
```

T.C = O(N)

S.C = O(1)

amp

$\wedge \equiv O(1)$ time

$\& \equiv O(1)$

$| \equiv O(1)$

Summary

1. Number System
2. Binary No System
3. Bitwise Operators + Properties
4. Questions $\equiv$ $\underline{\wedge}$ looks for exclusivity

---

Bitwise Operators

$<<$ left shift

$>>$ Right shift

int = 4 bytes

int = 2 bytes

$a = 8$

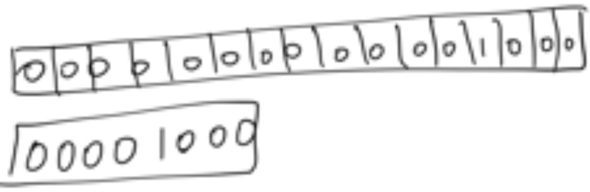
1000

int = 1 byte

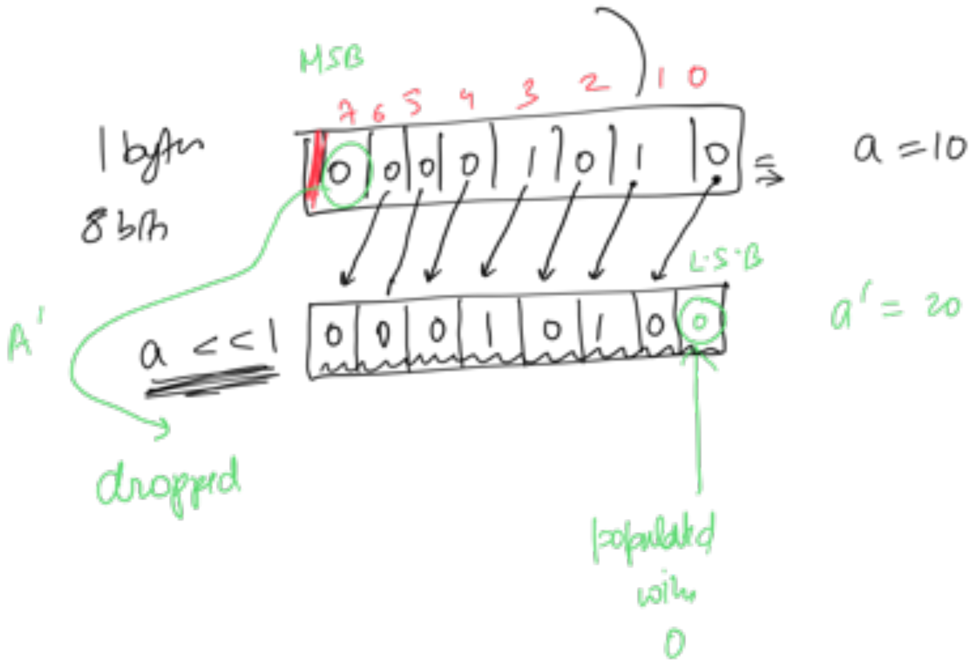
8 bits

Rejoining in 2 mins

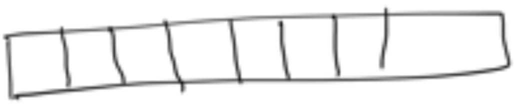
2 bytes = 16 bits

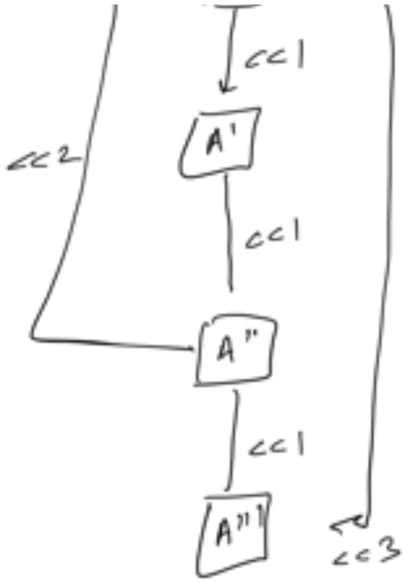


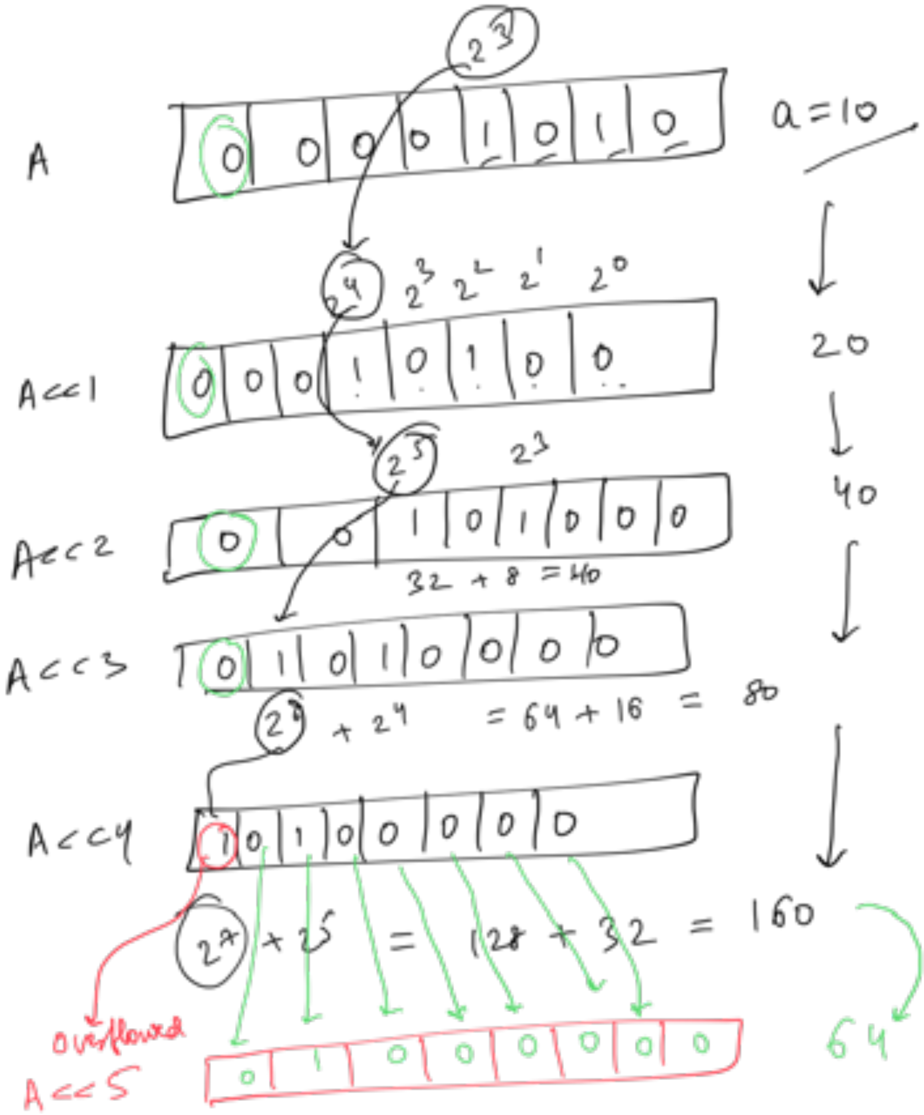
1000



a << 2







Till you have

at the most significant bit

The number will double on each $<<1$

Till overflow happens, the number will double

* assuming no overflow *

$$\boxed{a << n = a \times 2^n}$$

$$a << 1 = a \times 2$$

$$a << 2 = a \times 2 \times 2$$

$$a << n = \begin{cases} a \times 2^n & \text{if } a < 2^{[illegible]-n} - 1 \\ ? & \text{else} \end{cases}$$

$a = 1$

0	0	0	0	0	0	0	1

$$a << 1 = 2$$

$$a << 2 = 4$$

$$a << 3 = 8$$

$$\vdots$$

$$a << n = 2^n$$

assume no overflow

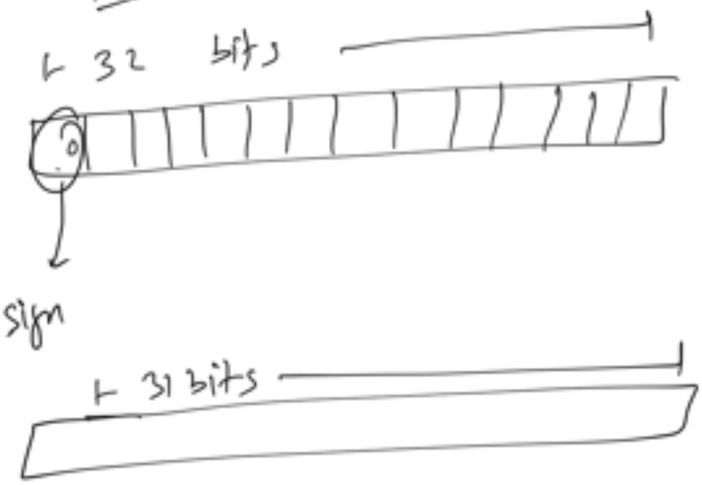
$$\boxed{1 << n = 2^n}$$

till no overflow

int = 4 bytes $\equiv$ 8 bits x 4

= 32 bits

$\boxed{\text{int32}}$ int32



$\boxed{2^{31}-1}$ = 2,147,483,648 − 1

32 bits

max number = $\boxed{2147483647}$

you can store in int32

10^9

<< 1

2×10^9

32 bit

int32 a = 2×10^9 + 10;

print(a x 2)

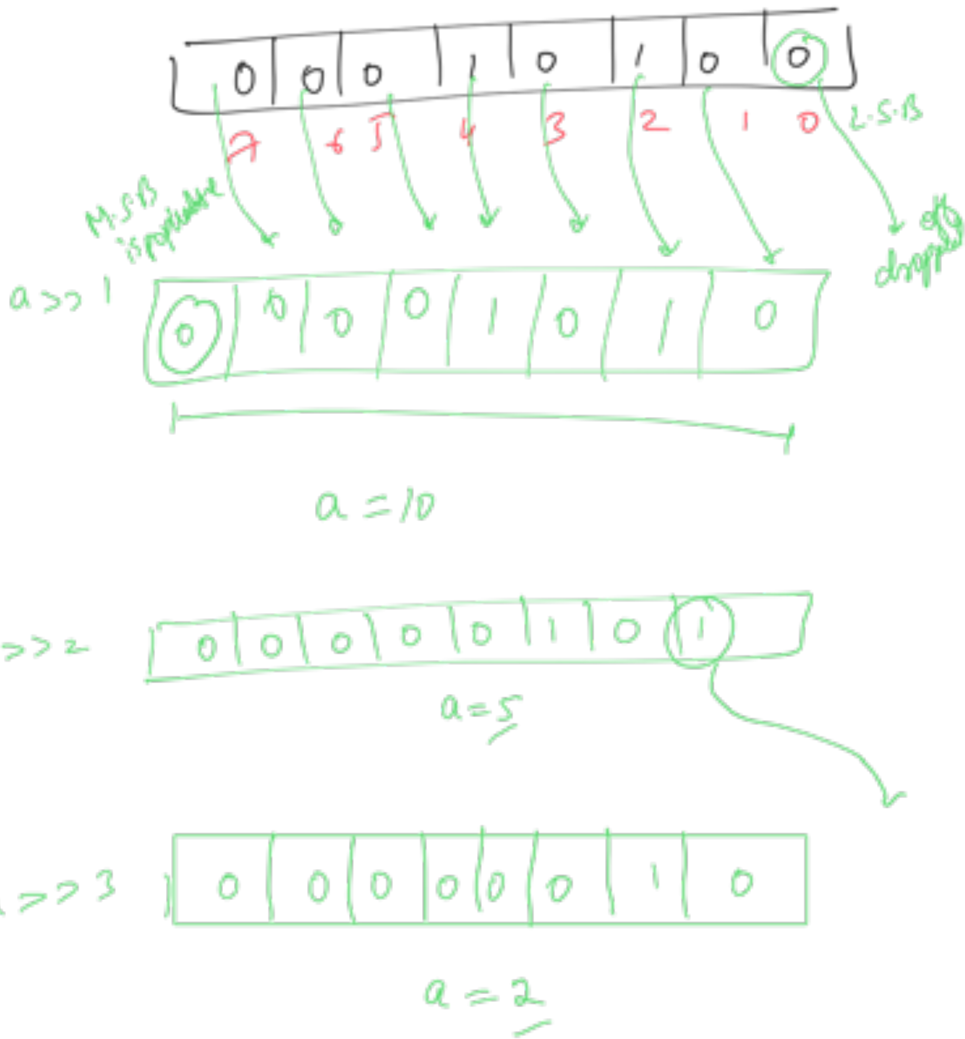
~~X~~ $4 \times 10^9 + 20$

= cheeky number you can't explain

Right shift

>>

a = 20



$a >> 4$ | 0 0 0 0 0 0 0 1 |

$a = 1$

$a >> 5$ | 0 0 0 0 0 0 0 0 |

$a = 0$

$a >> 6$ | 0 0 |

$a = 0$

$a >> 7$ | $a = 0$ |

$$a >> n = \frac{a}{2^n}$$

$$a >> 1 = \frac{a}{2}$$

$a >> 2 = \frac{a}{4}$

$a >> n = \frac{a}{2^n}$

$a = 1000$

$a \times \boxed{1024}$

2^{10}

$a \times 2^{10}$

$\boxed{a << 10}$

Quiz1

$a = 15 \xrightarrow{\times 2} 30 \xrightarrow{\times 2} 60$

$a << 2$

Quiz 2

$a = 29$

$a >> 2$

$28 \xrightarrow{/2} 14 \xrightarrow{/2} 7$

$2^n = \boxed{1 << n}$

Q4 5^n

No (a) $5 \times (1 << n)$ 5×2^n

No (b) $5 << n = 5 \times 2^n$

No (c) $5 << (n-1)$ $5 \times 2^{n-1}$

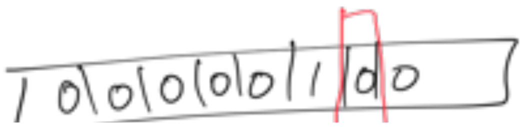
☆ (d) None of the above

Q the number N

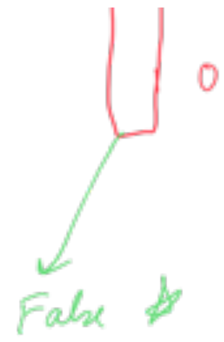
i

if the ith bit from the right is a set bit OR NOT

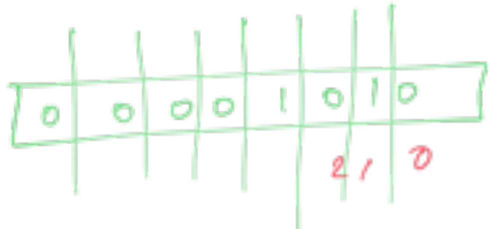
Ex => N=4



$i = 1$



$N = 10$

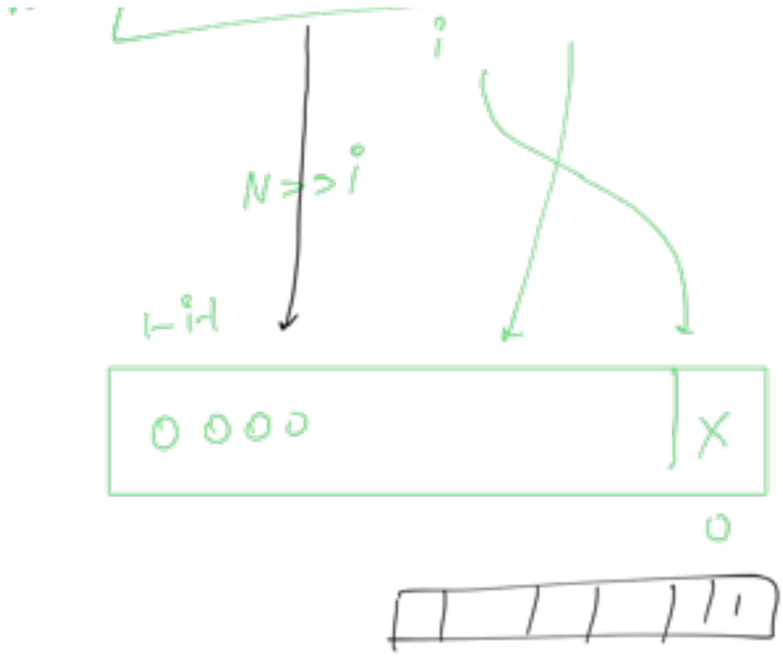


$i = 2$

False

$i = 3$ True

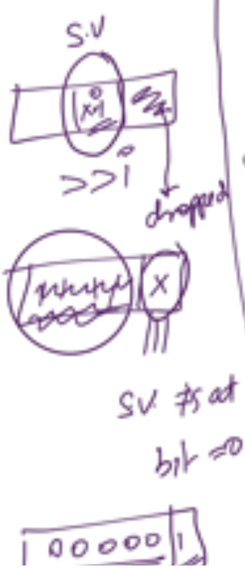
N



$(N >> i) \;\&\; 1$

True — ith bit was set

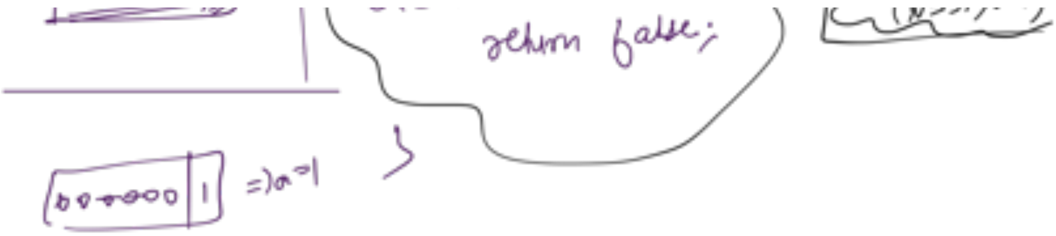
False — ith bit was unset

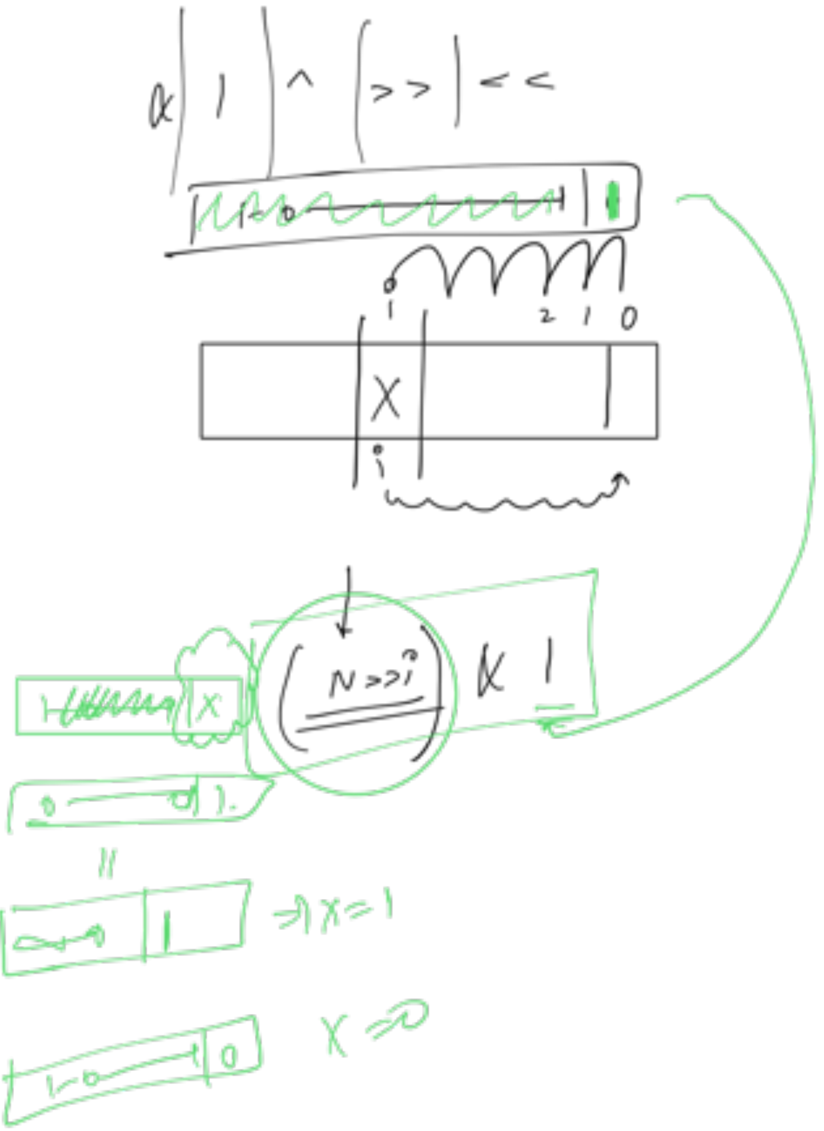


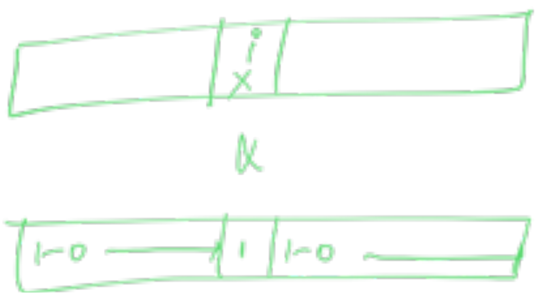

```
boolean checkBit (int N, int i)
{
    int N2 = N>>i;
    if (N2 & 1 == 1)
        return True;
    else
```

```
return N2&1;

return (N>>i)&1;
```


return false;
}
=> a=1


& | ^ >> <<
1
2 1 0
X
i
(N>>i) & 1
=> X=1
X = 0


i
X
&
1

7	6	5	4	3	2	1	0
0	1	1	1	0	0	0	1

N

k

0	0	0	1	0	0	0	0

B.N = beautiful number

if X = 1

0	0	0	1	0	0	0	0

$= 2^i = 1 << i$

else if X = 0

000 0 000

if ((N & B.N) == 0)
return false

else
return True

```
boolean checkBit( int N, int i)
{
    if (N & (1<<i)) == 0)
        return false
```

else

return true;

}

$T.C = O(1)$

aux $S.C = O(1)$

1<<i

N & 0 == 0

---

② Set the ith bit

int N

i

2 1 0

N=4 0 0 0 0 0 1 0 0

i=1

↓

N=4 with i=1 set = 00000 1 1 0

6

i = 0
N = 4

5

i = 2
N = 4

4

N = 0 1

N = 1 0 1 1 0 0 1 1 1 1

8 7 6 5 4 3 2 1 0

0 1 1 1 0 1 1 1

BITWISE OR

i = 5

B·N 0 0 0 1 0 0 0 0 0

If you want to make some specific bits = 0 or 1

You can create beautiful number accordingly

and that B.N will act as a mask,
and you can apply your mask to affect bits at the right place

Steps
1. Crafted Number ≡ B.N, that will allow you to identify those special index(es)
2. Operation that you need to perform b/w N and CN

Beautiful Number ≡ Crafted Number

```
int setBit(int N, int i)
{
    return N | (1<<i);
}
```

1 = | 0 0 0 0 0 | 1 |

1<<i | 0000 | 1 | 00000 |

Q

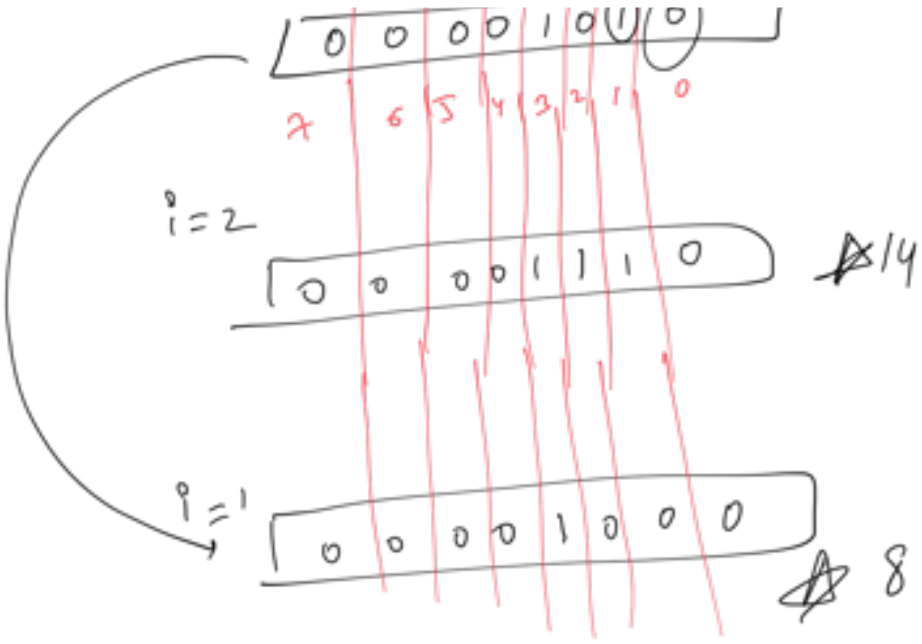
Amazon

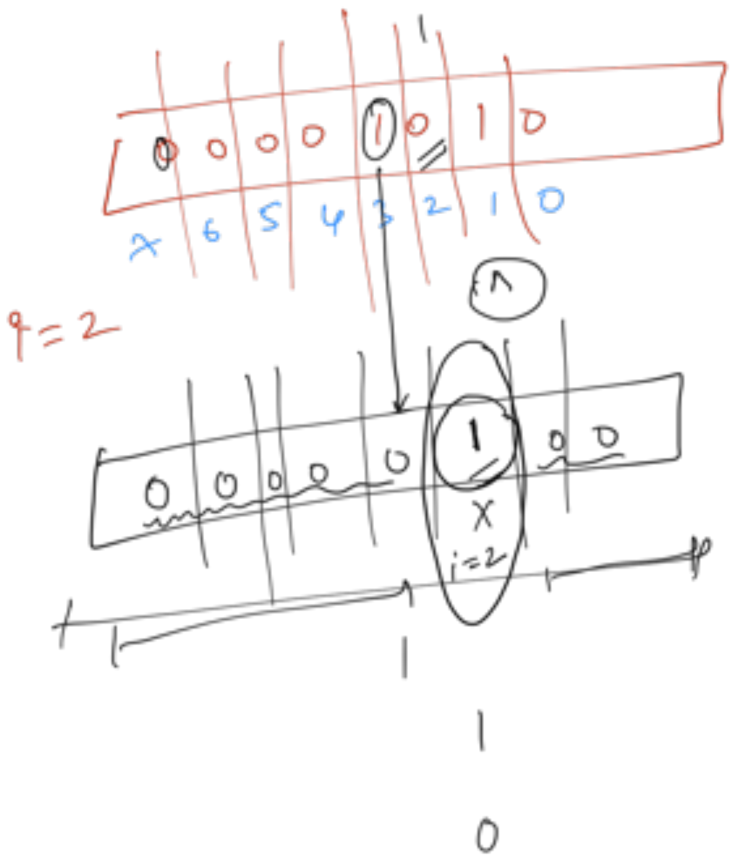
N

i

Toggle the ith bit for me

N=10





$Bit(X) \wedge 0 = X$

$Bit(X) \wedge 1 = \sim X$

```
int toggleBit(int N, int i)
{
    return N^(1<<i);
}
```

$0 \wedge 0 = 0$

$0 \wedge 1 = 1$

$1 \wedge 0 = 1$

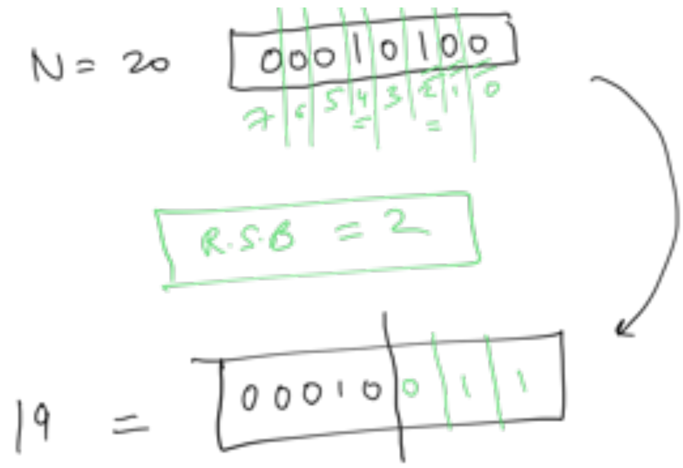
$1 \wedge 1 = 0$

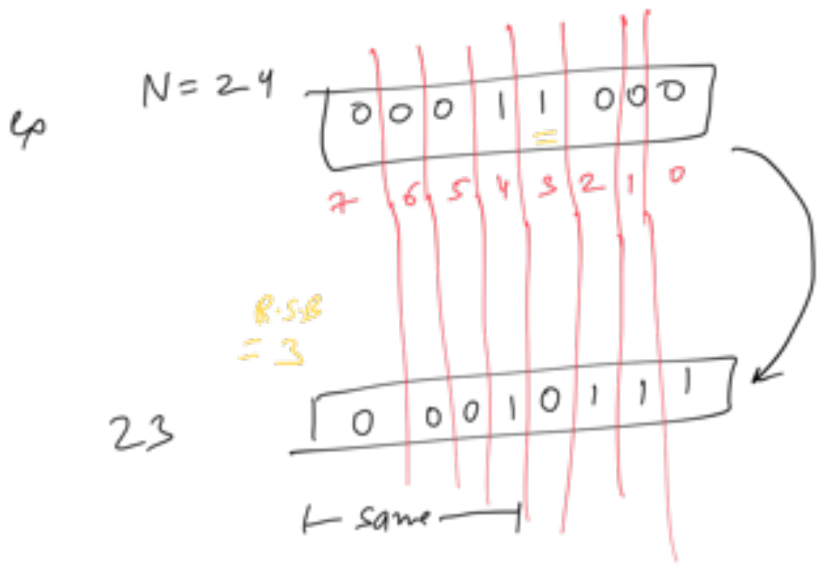
A=

Q

Tower Research

Given a +ve number N.
Toggle all the bits starting from the rightmost set bit.





int32

```
for (i= 0, i<=31, i++)
{
    //if NOTSET -> TOGGLE
    if (checkBit (N,i)== False)
        set(N, i);

    else
    {
        toggle(N, i);
        break;
    }
```

)

1111

RSBNF = True

i = 0

while (i <= 31 && RSBNotFound = True)

{

if (checkBit(N, i) == True)

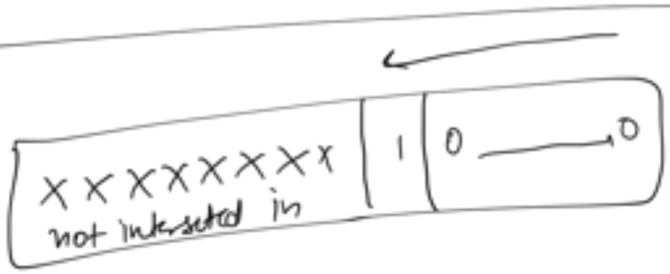
RSBNF = False

toggle(N, i)

}

$T.C = O(1)$

aux $S.C = O(1)$



obs (1) At the end you will find a

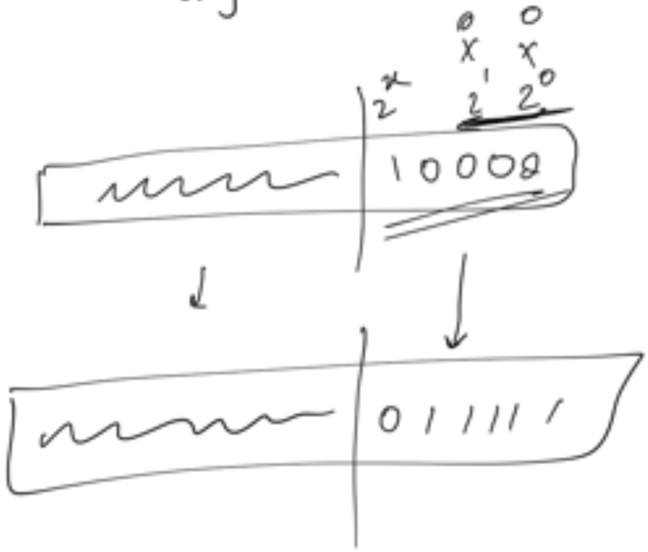
pattern

like

1 [0] k times

obs (2) To the left of this pattern,

nothing interests me



1 [0] k times

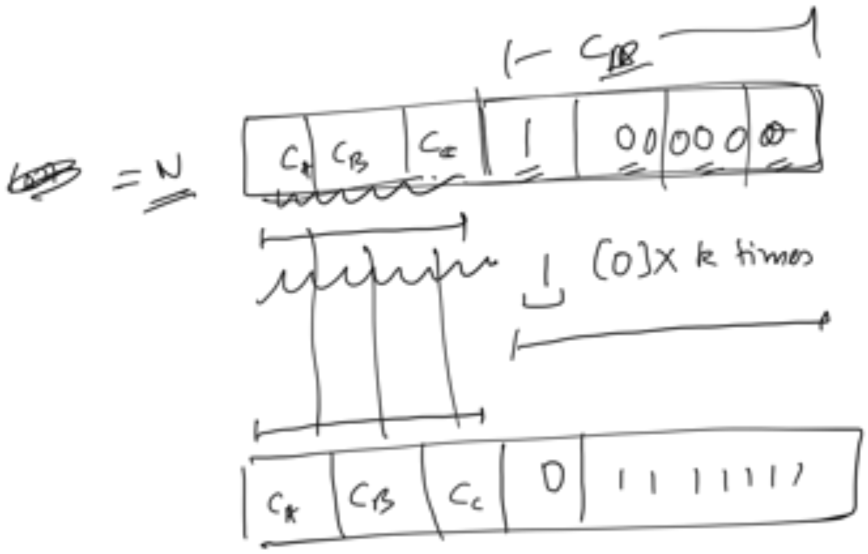
↓

0 [1] k times

2		1	
10	→	01	
4		3	
100	→	011	
1000	→	0111	
8		7	

N

$\cancel{A}\ N-1$



= (subtracts 1)

$$
\begin{array}{r}
0111111 \\
+\quad 1 \\
\hline
1000000
\end{array}
$$

$C_A + C_B + C_C + C_D - 1$

$N = C_A + C_B + C_C --- + C_D$

$N' = c_A + c_B - \text{———}$ B

$c_B - 1$

$\boxed{N' = N-1}$ ☺

$N = \boxed{2^{1000}}$ X X

$= O(\log_2 N)$

$2^{32} \sim 10^9$

$2^{64} \sim =$

int A